

Aether — Enterprise Secrets Management

AES-256-GCM Encryption, Rotation Policies, and Access Auditing

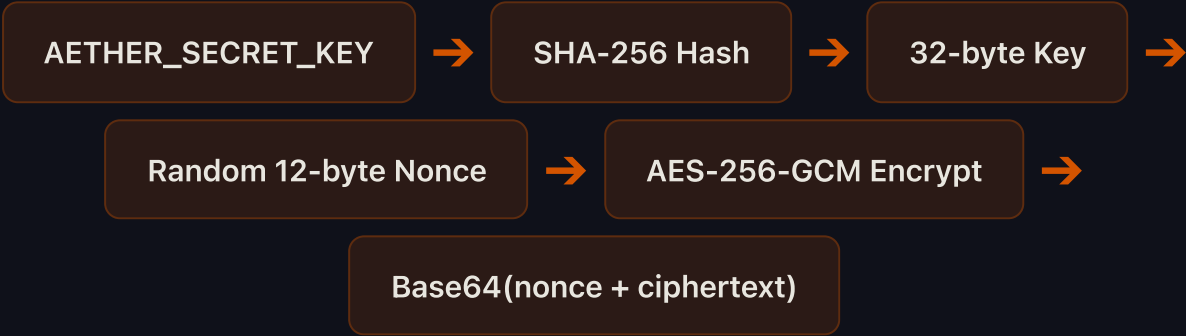
Create, encrypt, rotate, and audit secrets across all runtimes with a single secrets engine built into the control plane.

AES-256-GCM • SHA-256 Key Derivation • Random Nonces

AES-256-GCM Encryption

Every secret value is encrypted at rest using AES-256-GCM — the same authenticated encryption standard used by cloud providers and financial institutions.

Encryption Flow



Encryption Properties

Property	Value	Purpose
Algorithm	AES-256-GCM	Authenticated encryption with 128-bit auth tag
Key Derivation	SHA-256	Deterministic 32-byte key from any-length input
Nonce	12 bytes, cryptographically random	Unique per encryption, prevents replay attacks
Output Format	Base64(nonce + ciphertext)	Safe for JSON, YAML, environment variables
Tamper Detection	GCM authentication tag	Wrong key or corrupted data fails decryption

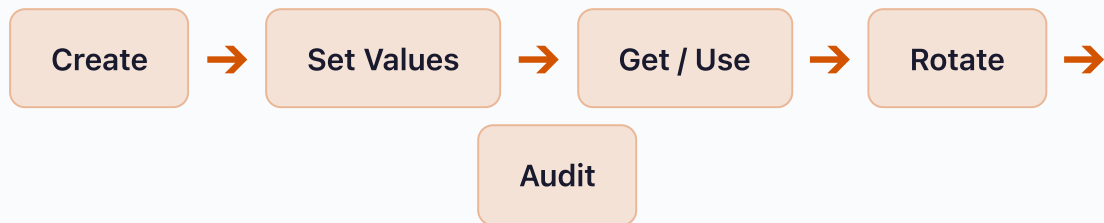
Key Management

The encryption key is loaded from `AETHER_SECRET_KEY` environment variable. If not set, Aether derives a machine-specific key from hostname using SHA-256. Production deployments should always set an explicit key of at least 16 bytes. Each encryption operation generates a fresh cryptographically random nonce, so encrypting the same value twice produces different ciphertext.

AES-256-GCM provides both confidentiality and authenticity — any tampering with the ciphertext is detected on decryption, and a wrong key produces a clear error rather than garbage output.

Secret Lifecycle

Manage secrets through their complete lifecycle: create, set, get, rotate, and audit. Every operation is tracked with actor identity and timestamp.



Lifecycle Commands

```
# 1. Create a secret container
$ aether secrets create db-creds --namespace production

# 2. Set encrypted values
$ aether secrets set db-creds password "s3cure-p@ssw0rd"
$ aether secrets set db-creds username "app-user"

Key 'password' set in 'db-creds' (version 1, AES-256-GCM)
Key 'username' set in 'db-creds' (version 1, AES-256-GCM)

# 3. Retrieve decrypted values
$ aether secrets get db-creds password

s3cure-p@ssw0rd

# 4. Rotate a value
$ aether secrets rotate db-creds password "n3w-s3cure-p@ss"

Key 'password' rotated in 'db-creds' (version 2)

# 5. View access audit trail
$ aether secrets audit db-creds
```

Version Tracking



Auto-Versioning

Every `set` or `rotate` operation increments the key's version counter. This provides a clear history of how many times a secret has been updated without storing old values.



Last Rotated Timestamp

Each secret value records when it was last rotated as an RFC 3339 timestamp. This timestamp drives the rotation policy engine and expiry alerting system.

Rotation Policies

Enforce automatic rotation schedules and expiry alerts. Every secret is created with a default rotation policy that can be customized per use case.

Default Rotation Policy

Parameter	Default Value	Description
<code>interval_days</code>	90	Rotate every 90 days for compliance
<code>max_age_days</code>	365	Force rotation after 1 year regardless
<code>notify_before_days</code>	14	Alert 14 days before max age expiry

Rotation Audit

```
$ aether secrets rotation-audit
```

```
[CRITICAL] Key 'api-key' in 'stripe-creds' is 400 days old (max: 365)
```

```
[WARNING] Key 'password' in 'db-creds' expires in 12 days
```

```
[OK] Key 'token' in 'github-creds' rotated 15 days ago (next: 75 days)
```

Policy Validation

Rotation policies are validated on creation to prevent misconfiguration:

- `interval_days` must be greater than 0
- `max_age_days` must be greater than or equal to `interval_days`
- `notify_before_days` must be less than or equal to `max_age_days`

This prevents impossible policies like "notify 400 days before a 365-day expiry" that would cause constant false-positive alerts.



Warning Alerts

When a secret's age exceeds `max_age_days` - `notify_before_days`, Aether emits a WARNING alert. This gives teams a 14-day window (by default) to plan rotation.



Critical Alerts

When a secret exceeds `max_age_days`, the alert escalates to CRITICAL. The secret is overdue for rotation and may violate compliance requirements.

Access Audit Trail

Every secret operation is logged with actor identity, action type, key name, and timestamp. The audit trail provides complete visibility into who accessed what and when.

Tracked Actions

Action	Trigger	Logged Data
READ	aether secrets get	Actor, secret name, key, timestamp
WRITE	aether secrets set	Actor, secret name, key, version, timestamp
DELETE	aether secrets delete-key	Actor, secret name, key, timestamp
ROTATE	aether secrets rotate	Actor, secret name, key, new version, timestamp

```
$ aether secrets audit db-creds
```

```
Secret Access Log: db-creds
```

Timestamp	Actor	Action	Key
2026-04-10T08:00:00Z	admin	WRITE	password (v1)
2026-04-10T08:00:05Z	admin	WRITE	username (v1)
2026-04-11T14:30:00Z	deploy-bot	READ	password
2026-04-11T14:30:01Z	deploy-bot	READ	username
2026-04-12T09:00:00Z	admin	ROTATE	password (v2)
2026-04-12T10:00:00Z	admin	DELETE	old-key

CLI Commands Reference

Secret Management

- `aether secrets create <name>`
`--namespace <ns>`
- `aether secrets set <name>`
`<key> <value>`
- `aether secrets get <name>`
`<key>`
- `aether secrets delete-key`
`<name> <key>`
- `aether secrets delete <name>`

Rotation & Auditing

- `aether secrets rotate <name>`
`<key> <value>`
- `aether secrets rotation-`
`audit`
- `aether secrets audit <name>`
- `aether secrets list`
- `aether secrets list --needs-`
`rotation`

Secure Your **Secrets**

Enterprise-grade secrets management built into every Aether deployment — no external vault required.

256

Bit Encryption Key

4

Audited Actions

90d

Default Rotation Interval

Secrets That Manage Themselves

AES-256-GCM encryption, automatic rotation policies, expiry alerts, and complete access auditing — for every secret across every runtime.

```
# Complete secrets workflow
$ export AETHER_SECRET_KEY="$(openssl rand -hex 32)"
$ aether secrets create app-secrets --namespace production
$ aether secrets set app-secrets db-password "s3cure-v@lue"
$ aether secrets rotate app-secrets db-password "n3w-v@lue"
$ aether secrets rotation-audit
$ aether secrets audit app-secrets
```

Aether — Universal Runtime Control Plane